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❖ REPORT

TOPIC - An 8051-Based Secure Offline Wallet

COURSE CODE :- ECE322

Submitted to :- Ch.Ravi Sankar

Submitted by:-

Sarabjit Yadav -12408908-(12)

Piyush Prasad -12404205-(03)

Akash Shaw -12416333-(17)

Deepak Kumar Mohapatra -12408830-(02)

An 8051-Based Secure Offline Wallet

Abstract

The “8051-Based Secure Offline Wallet” is an embedded system project developed using the AT89S52 Microcontroller to provide secure offline banking operations without requiring internet connectivity. The system uses a 4x4 keypad for user input, a 16x2 LCD for displaying information, and an IR sensor for automatic user detection. When a person stands in front of the device, the IR sensor activates the system and the LCD prompts the user to enter a secure PIN. After successful authentication, the user can check account balance, transfer balance to another user, and withdraw money. All transaction details and messages are displayed on the LCD screen. The project is implemented using Assembly Language programming in the Keil uVision IDE environment. This system demonstrates the practical use of microcontrollers in secure banking and authentication applications.

Introduction

With the rapid growth of digital banking systems, secure and reliable transaction methods are becoming increasingly important. Most modern banking systems depend heavily on internet connectivity, which may not always be available in rural or remote areas. Therefore, there is a need for a low-cost and secure offline transaction system.

The “8051-Based Secure Offline Wallet” is designed to perform basic banking operations using an embedded system without internet access. The project is based on the AT89S52 Microcontroller and provides password-protected access to wallet functionalities. An IR sensor is used to detect the presence of a user, making the system energy-efficient by turning the LCD ON only when needed.

The system combines security, automation, and user interaction using simple electronic components and Assembly Language programming.

Problem Statement

Traditional digital payment systems require internet connectivity and advanced hardware, which increases system complexity and cost. In many situations, users need a simple and secure offline method to perform basic banking operations such as balance checking and fund transfer.

The main problems addressed in this project are:

- Lack of secure offline transaction systems
- High dependency on internet-based banking
- Need for low-cost embedded banking solutions
- Requirement of user authentication for secure access
- Efficient power usage in embedded systems

This project aims to solve these issues using an 8051 microcontroller-based embedded system.

Components Used

Component	Quantity	Purpose
AT89S52 Microcontroller	1	Main controller of the system
4x4 Keypad	1	User input for PIN and operations
16x2 LCD Display	1	Displays balance and transaction
MB102 Power Module	1	Power supply unit
Breadboard and Jumper Wires	1 Kit	Circuit connections
11.0592 MHz Crystal Oscillator	1	Clock signal generation
22pF Capacitors	2	Crystal stabilization
0.1μF Capacitors	2–3	Noise filtering

Component	Quantity	Purpose
10μF Capacitor	1	Power stabilization
10k Resistors	2	Pull-up and reset circuit
4.7k Resistors	2	Signal conditioning
10k Potentiometer	1	LCD contrast adjustment
Push Button	1	System reset
IR Sensor	1	User detection and automatic LCD activation

Objectives

The major objectives of this project are:

1. To design a secure offline wallet system using the 8051 microcontroller.
2. To implement password-based authentication using a keypad.
3. To detect user presence automatically using an IR sensor.
4. To display banking operations on a 16x2 LCD screen.
5. To perform balance checking, withdrawal, and balance transfer operations.
6. To develop the software using Assembly Language in the Keil uVision IDE.
7. To create a low-cost and efficient embedded banking prototype.

Implementation

Hardware Implementation

The hardware circuit is built around the AT89S52 Microcontroller. The 11.0592 MHz crystal oscillator provides the clock frequency required for microcontroller operation. Capacitors and resistors are connected for proper reset and signal stabilization.

The 4x4 keypad is interfaced with the microcontroller for entering PINs and selecting banking operations. The 16x2 LCD is connected in parallel mode to display all system messages and transaction details.

The IR sensor continuously monitors user presence. When a person stands in front of the system, the sensor output activates the LCD and starts the authentication process. After successful PIN verification, the user can:

- Check balance
- Withdraw balance
- Transfer balance
- View transaction confirmation

A push button is included for resetting the system whenever required.

Software Implementation

The software part of the project is developed using Assembly Language in the Keil uVision IDE.

The software performs the following tasks:

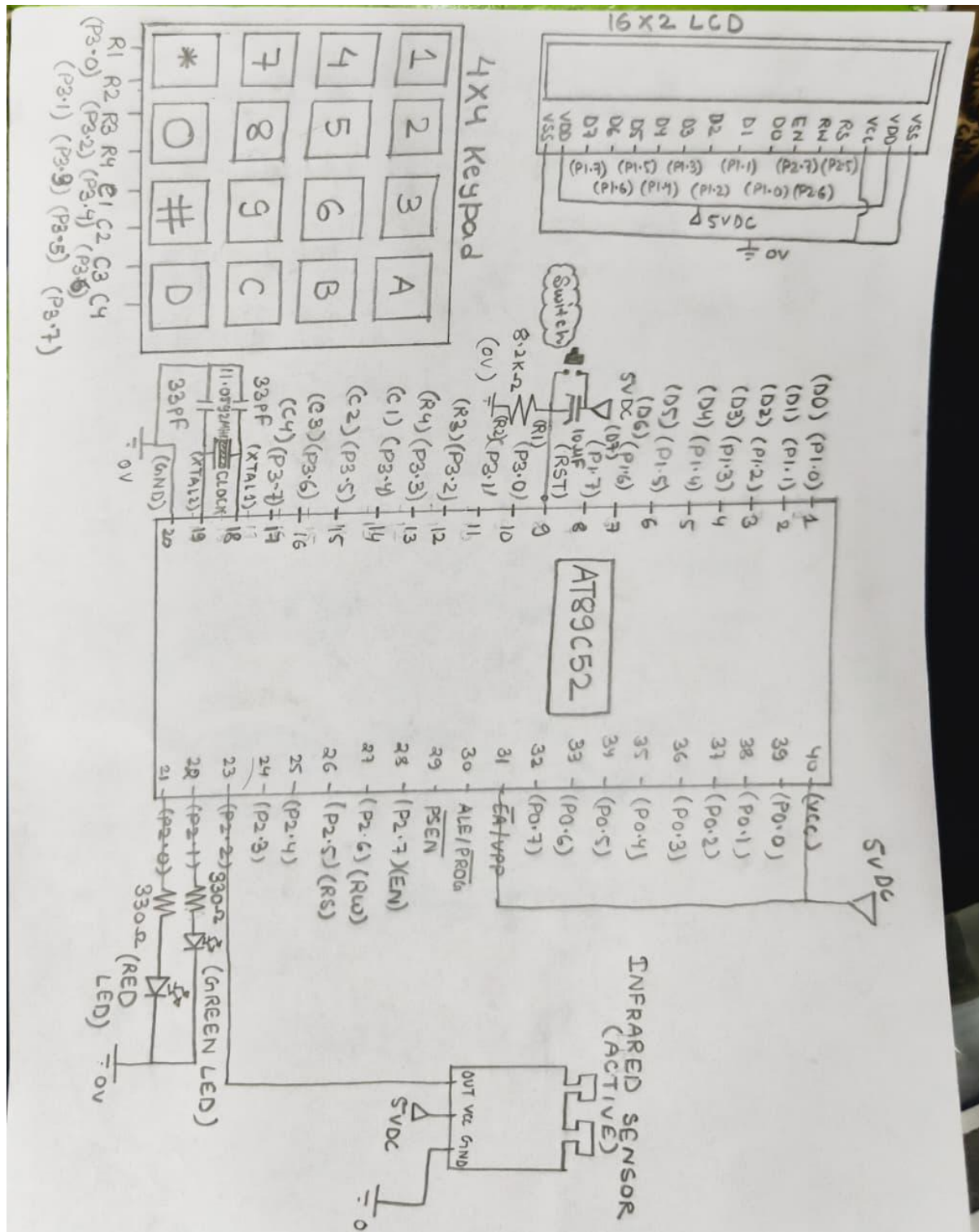
1. Initialization of LCD, keypad, and ports
2. Reading sensor input from the IR sensor
3. Displaying welcome and PIN messages
4. Verifying the entered password
5. Processing banking operations
6. Updating balance values
7. Displaying transaction status on LCD

The program flow follows these steps:

1. IR sensor detects user
2. LCD turns ON
3. User enters PIN
4. PIN verification occurs

5. Menu options appear
6. User selects banking operation
7. LCD displays results

CIRCUIT DIAGRAM



Result Analysis

The developed system successfully performs secure offline banking operations using the 8051 microcontroller. The IR sensor correctly detects user presence and activates the display system automatically. Password authentication ensures that unauthorized users cannot access wallet functions.

The keypad interface allows smooth user interaction, while the LCD clearly displays all transaction information. The system successfully demonstrates balance checking, money withdrawal, and balance transfer operations in real time.

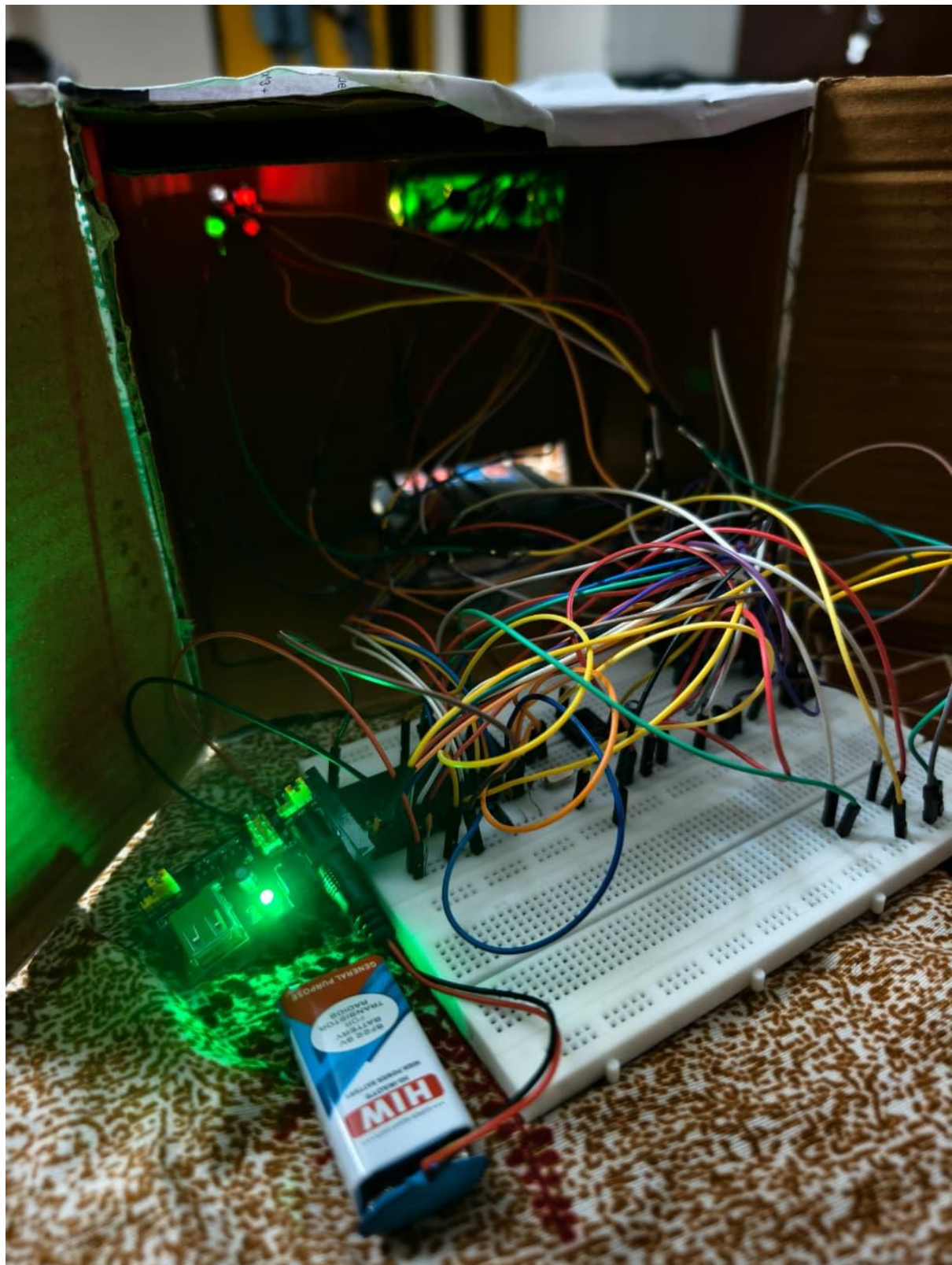
The project proves that embedded systems can be effectively used for secure offline financial applications with low hardware cost and simple programming techniques.

Applications

The “8051-Based Secure Offline Wallet” can be used in several applications such as:

- Offline banking systems
- Secure digital wallet prototypes
- Smart ATM learning systems
- Educational embedded system projects
- Rural banking solutions
- Authentication-based transaction systems
- Embedded security applications





Project Development Schedule

Week 1: Hardware circuit testing and design

Part 1: Power Supply and Microcontroller Setup

Hours Worked:

- **4 Hours**

Work Done:

- Connected the AT89S52 Microcontroller on breadboard
- Designed power supply section using MB102 module
- Connected reset circuit and crystal oscillator
- Tested microcontroller power stability
- Verified clock signal operation

Part 2: LCD and IR Sensor Interfacing

Hours Worked:

- **5 Hours**

Work Done:

- Interfaced 16x2 LCD with the microcontroller
- Adjusted LCD contrast using potentiometer
- Connected IR sensor for user detection
- Programmed automatic LCD activation
- Tested sensor response and LCD display

Week 2

Part 3: Keypad Interfacing and PIN Entry System

Hours Worked:

- **3 Hours**

Work Done:

- Connected 4x4 keypad with controller ports
- Developed keypad scanning logic
- Programmed PIN entry system
- Displayed entered values on LCD
- Tested keypad response accuracy

Part 4: Authentication and Banking Menu Display

Hours Worked:

- **6 Hours**

Work Done:

- Implemented password verification logic
- Added wrong PIN detection
- Created menu options for:
 - Balance Check
 - Withdraw Balance
 - Transfer Balance
- Programmed LCD menu navigation
- Tested complete login system

Week 3

Part 5: Balance Management and Transaction Operations

Hours Worked:

- **5 Hours**

Work Done:

- Developed balance calculation logic
- Added withdrawal operation
- Added balance transfer feature
- Updated balance after transactions
- Displayed transaction confirmation messages on LCD

Part 6: Final Integration, Testing, and Debugging

Hours Worked:

- **4 Hours**

Work Done:

- Integrated all hardware and software modules
- Debugged Assembly Language program in Keil uVision IDE
- Corrected LCD display and keypad errors
- Tested IR sensor operation
- Performed final project testing and verification

Summary Table

Week	Part No.	Work Description	Hours Worked
Week 1	Part 1	Power Supply and Microcontroller Setup	4 Hours
Week 1	Part 2	LCD and IR Sensor Interfacing	5 Hours
Week 2	Part 3	Keypad Interfacing and PIN Entry System	3 Hours
Week 2	Part 4	Authentication and Banking Menu Display	6 Hours
Week 3	Part 5	Balance Management and Transaction Operations	5 Hours
Week 3	Part 6	Final Integration, Testing, and Debugging	4 Hours

Total Working Hours: 27 Hours

Conclusion

The “8051-Based Secure Offline Wallet” is a secure and efficient embedded banking system developed using the AT89S52 Microcontroller. The project successfully integrates password authentication, IR-based user detection, keypad input, and LCD display functionality to perform offline wallet operations.

The system provides a low-cost solution for secure financial transactions without requiring internet connectivity. This project also demonstrates the practical implementation of Assembly Language programming and hardware interfacing techniques in embedded systems.